

WRRRA Core 1.0

A Domain-Agnostic Execution Architecture for the Wonsik Reality- Renderer Architecture

Domain-Agnostic Execution Architecture

Core Specification Preserving Minimum Computation, the Common Carrier, and Phenotype

Freeze Declaration

WRRRA Core 1.0 is frozen as a domain-agnostic execution grammar that is not dependent on the equations or objects of any particular field, including physics, biology, artificial intelligence, economics, meteorology, mathematics, or methodology. Later revisions to Domain Profiles such as MCC/WRRRA-Physics, WRRRA-Bio, WRRRA-AI, WRRRA-Econ, and WRRRA-Met do not automatically change Core 1.0. The Core changes only when one of the constitutional invariants or required execution elements defined below is itself revised.

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0. WRR Core 1.0 in One Sentence

WRR Core 1.0

In any system, SOURCE produces the current state through relations and boundaries; effects of the past operate only through RESIDUE/RECORD remaining in the present; multiple channels reuse a common carrier wherever possible; the next present is updated with the minimum computation required; the RENDERER outputs an observable PHENOTYPE; and the outcome remains as an OBSERVABLE/LEDGER/RECORD.

$$\begin{aligned} \text{SOURCE} \rightarrow \text{RELATION/LAW} \rightarrow \text{STATE/RESIDUE} \rightarrow \text{BOUNDARY} \rightarrow \text{COMMON} & \quad (0.1) \\ \text{CARRIER} \rightarrow \text{UPDATE} \rightarrow \text{RENDERER} \rightarrow \text{PHENOTYPE} \rightarrow \\ \text{OBSERVABLE/RECORD/LEDGER} \end{aligned}$$

This sequence generalizes the SOURCE-to-Reality model developed in physics. In some domains, LAW may be specified as a separate layer; in others, it may be incorporated into RELATION/UPDATE rules. OBSERVABLE, RECORD, and LEDGER may likewise be separated or combined in an implementation. Nevertheless, the three constitutional axes—minimum computation, the common carrier, and phenotype—and the fixed-present principle that execution proceeds through the current state are retained in Core 1.0.

1. Three Constitutional Invariants

1.1 Minimum Computation — Minimum Viable Computation

WRR Core 1.0 does not aim for peak performance, maximum information, maximum memory, or maximum complexity. Its central rule is to compute only as much as is needed to cross the declared boundary of expression, function, survival, or stability, without adding unnecessary states, relations, memory, transport, or reserve beyond that point. Reserve or redundancy required for viability under extreme environments, however, is not waste; it belongs to the minimum cost of survival.

$$A^* \in \text{ParetoMin} \{ C(A) \mid A \text{ satisfies declared viability / expression contract} \} \quad (1.1)$$

Prohibited Misreading

Minimum computation does not mean “compute as little as possible” without qualification. If required distinctions are removed until expression or survival collapses, the result is deficiency, not minimality. Conversely, continuing to add relations, memory, carriers, or resources after the boundary has been crossed is excess.

1.2 Common Carrier

When different channels do not require separate origins and separate transmission mechanisms, WRR Core 1.0 uses one or a small number of reusable transport/update structures as a common carrier. A common carrier does not erase distinctions among channels. Different cargoes can retain distinct ownership and destinations even when they travel on the same road.

$$D_C : \oplus_i H_i \rightarrow \oplus_i H_i, \quad \text{shared transport} \neq \text{erased channel identity} \quad (1.2)$$

The common carrier is not limited to the 15-channel carrier of physics. In biology, it may be a reusable execution scaffold integrating translation, metabolism, membrane transport, and partitioning; in AI, a sensory-state-action transmission network; in economics, a transaction, price, credit, and information network; and in meteorology, a shared update field for atmospheric state and flow. Each Domain Profile must state what is transported and which common structure is reused.

1.3 Phenotype

WRR Core does not assume that SOURCE or internal STATE is identical to the object of observation. Internal information, states, and relations pass through a RENDERER and a BOUNDARY to become a phenotype that can actually be read. Phenotypes are domain-specific: particles, forces, and vacuum output in physics; proteins, cellular states, and differentiation in biology; actions in AI; prices, transactions, and crisis states in economics; and observable tropical-cyclone tracks in meteorology.

$$\text{PHENOTYPE}_t = \Phi_D(\text{STATE}_t, \text{BOUNDARY}_t, \text{RENDERER}_D) \quad (1.3)$$

Core Principle

The shortcuts “SOURCE is the outcome,” “internal state is observation,” and “what is possible is what is actually occupied” are prohibited. A Domain Profile must explain the renderer between internal state and phenotype.

2. Required Execution Elements of Core 1.0

Core Element	Audit Question	Example
SOURCE	What independent information, resources, and boundary inputs enter current execution?	DNA / sensor / market shock / initial field
RELATION / LAW	What affects what, and which rules are permitted?	coupling / network / physical law / reaction
STATE	What internal state actually exists now?	concentration / activity / field / price state
RESIDUE / CARRY	What effect of the past remains in the present?	memory / adaptation / epigenetics / debt / environmental residue
BOUNDARY	What lies inside or outside the system, and which flows open or close?	membrane / body / market institution / observation region
COMMON CARRIER	What transport/update scaffold is shared across channels?	transport operator / translation system / network
UPDATE / TAKEOVER / RESET	How does the current present become the next present?	dynamics / division / policy / state transition
RENDERER	By what rule does an internal state become an external expression?	measurement channel / protein execution / action policy

Core Element	Audit Question	Example
PHENOTYPE	What output/state is actually instantiated?	particle / cell fate / action / price / cyclone state
OBSERVABLE	What can actually be measured?	sensor value / omics / survival / track error
RECORD / LEDGER	What is persistently recorded, and which totals/responsibilities are tracked?	history / resources / energy / generation record

2.1 Fixed-Present Principle

WRRR Core does not require a structure that directly rereads the entire sequence of past events at every step. If the past affects the present, its effect must remain physically or informationally instantiated in the current STATE, RESIDUE, or RECORD.

$$X_{t+1} = F_D(X_t, S_t, R_t, B_t, C_t), \quad \text{history enters only through present-instantiated residue/record} \quad (2.1)$$

This does not mean that memory is absent. Rather, it fixes the ownership of memory in the present. A model may refer directly to a past log, but it must specify that the log is a RECORD accessible in the present.

2.2 Ownership Principle

Even the same number, signal, or state is not the same variable when it belongs to a different owner. WRRR specifies the owner of information, resources, forces, memory, and outputs. This principle makes it possible to distinguish gravitational $T_{\mu\nu}$ from gauge currents in physics, DNA from translation machinery in biology, sensors from memory in AI, and current sentiment from genuine residue in economics.

$$\text{Owner}(z) \text{ must be declared before } z \text{ is used as state, residue, carrier, or observable} \quad (2.2)$$

2.3 Fail-Closed Boundary Principle

WRRR does not inflate structural executability into empirical validation by nature. Interpretation, computation, prediction, control, and empirical validation are distinct contracts. When data, time, causality, or comparison baselines are insufficient, the result is frozen as OPEN or PARTIAL.

CONTRACT	Core Definition
INTERPRETATION	Reconstruct a current or already recorded target within tolerance using current/recorded information
COMPUTATION	Produce the next state from a complete current state under a given update rule
PREDICTION	Reduce the loss or uncertainty distribution of a future target using current information
CONTROL	Select actions that change future viability or objectives

CONTRACT	Core Definition
OPEN	The contract is not closed with respect to information, error, time, causality, or computational budget

3. Separation of Core and Domain Profiles

WRRA Core 1.0 contains no domain-specific equations or objects. Each application is managed as a Domain Profile. A Domain Profile maps the actual owners and equations of Core elements, but changes in that profile do not automatically change the Core version.

$$\text{WRRA Domain Profile } D = \text{Map_D}(\text{Core elements} \rightarrow \text{domain states, carriers, renderers, observables, ledgers}) \quad (3.1)$$

Verdict	Example of Change
Core 1.0 retained	Domain-specific coefficients, equations, datasets, or carrier implementations change
Core 1.0 retained	MCC/WRRA-Physics is updated from 2.0 to 2.1 to 2.2
Core 1.0 retained	Experimental results in WRRA-Bio/AI/Econ/Met change from success to failure, or vice versa
Review Core major/minor	The definition of minimum computation itself changes
Review Core major/minor	The common carrier or phenotype is removed from or redefined within the required axes
Review Core major/minor	The fixed-present, ownership, or fail-closed principle is abandoned

Version Rule

Future notation is separated as follows: WRRA Core 1.0 / WRRA-Physics (MCC) 2.x / WRRA-Bio 4.x / WRRA-Cell 0.x / WRRA-AI 2.x / WRRA-Worm 0.x–1.x / WRRA-Econ 0.x / WRRA-Met 0.x / WRRA-Method 2.x. Reinforcing physics with string theory or incrementing the cosmology version does not require revisions to documents in biology, AI, economics, or meteorology.

4. Official Application Lineage — Through 7 September 2026

The following registry reclassifies, from the perspective of Core 1.0, the applications that had been developed as models, papers, or validation programs using the WRRR name and execution grammar by the freeze date. “Application” does not mean that the Core has been proven correct in that field; it means that the Core grammar has been implemented as a particular Domain Profile.

4.1 WRRR-Physics / Minimal Computing Cosmology

Application / Research Line	Core Mapping	Meaning of the Freeze
WRRR Axiomatic / Physical Realization	Separates the layers of classical, relativistic, and quantum possibility from physical realization	Upper-level physics architecture
Minimal Computing Cosmology 2.0	LAW→SOURCE→EXECUTION→RENDERER→REALITY→RECORD; minimum dimension, structure, and computation	Physics/cosmology reference version
MCC 2.1 string-microphysics repair	Retains the MCC skeleton and uses string theory as microphysical repair material	Physics Profile reinforced; Core unchanged
Dimensional filter / 3+1 Reality	Distinguishes physical spacetime from internal, Hilbert-space, and registry dimensions	Structural minimality
Asymmetric unification of the four forces	Total gravitational $T_{\mu\nu}$ owner versus selective gauge-current owners	Theoretical result
Common carrier / 15 chiral channels	Common transport operator with selective expression	Central Physics carrier
Big Bang / opening	Dormant state, minimum SOURCE η , and a constraint-preserving boundary transition	Implementation of cosmic history
Hot Big Bang closure	Kination/energy transfer $Q_r \rightarrow$ radiation domination	Final reinforcement of 2.0
Matter–antimatter / matter-dominant Reality	Separates lawful sector from occupied Reality; tracks the owner of asymmetry	Dynamical/predictive target
Vacuum	Zero real-particle-output phenotype in a stationary background	Physical phenotype
Pair creation and annihilation	Conditional output and conservative transfer of carrier ownership	Repositioning of physical phenomena
Antimatter-generation scenario	Direction-history and pair-plasma validation protocol	Independent deviation OPEN
Dark matter	Conditions for a gauge-singlet candidate that remains in $T_{\mu\nu}$	Candidate/owner conditions
Black holes	Finite boundary information, internal transport/record, and compatibility fiber	Boundary/information application
Temperature and entropy	Accessible state count and multiplicity of histories indistinguishable from the current record	Coarse-graining application
Quantum–macro boundary / two-pass outcome	Lawful support, quantum instrument, record redundancy, and phenotype threshold	Method + physics
Superconductivity and superfluidity	Pairing/phase stiffness and boundary/carry/reset ledger	Structural reinterpretation; new surplus prediction OPEN

Application / Research Line	Core Mapping	Meaning of the Freeze
Nuclear decay / multiple decay clocks	Channel-specific barriers and transition rules separated from half-life clocks	Physics series
Stability of superheavy nuclei	Audit against fission/decay baselines with failure preservation	Validation-oriented application
Pulsars/intermittent pulsars and time correction	Application through rhythm, state transitions, records, and clocks	Physics extension
Macroscopic determinacy and microscopically open events	Boundary between probabilistic events and stable records	Physics methodology

4.2 WRRR-Math / Arithmetic and Proof-Boundary Research Line

Application	Core Mapping	Status
WRRR 2.0 prime/composite SOURCE-RELATION registry	Models independent SOURCE and folded relation addresses arithmetically	Historical upstream branch; not the Core
WJNS arithmetic stability gate	Uses zeta/Mellin/reflection structure as a stability gate	Separated from physical time
Riemann Hypothesis boundary studies	Distinguishes finite computation/verification from proof closure over an infinite totality	RH proof OPEN
RH audit by the Interpretation-Prediction classifier	Classifies a static infinite proposition as a proof-closure problem, not future prediction	Methodological application

This arithmetic research line does not justify placing primes, the zeta function, or the Riemann Hypothesis inside WRRR Core. It is a particular Domain Profile or a historical upstream implementation; WRRR applications in other domains are not required to use it.

4.3 WRRR-Bio / Life, Genetics, and Cells

Model / Application	Core Mapping	Current Meaning
Life Model 0 — First Living System	Defines the first living lineage when information, catalysis, boundary, energy, and partitioning produce generational closure with $R_{life} > 1$	Executable inference; historical first sequence/path OPEN
Life Model I — Minimal Modern Cell	Information, translation/execution, metabolism, membrane, and division/generation gate	Computational closure PASS; fully autonomous experiment OPEN
Life Model II — DNA→RNA→Protein	DNA as SOURCE; polymerase/ribosome as material renderer/executor; coding lower bound and bootstrap	Equations/structure EXACT; autonomy OPEN
Life Model III — Daughter/Lineage Closure	Finite molecular partitioning, active equalization, and generational reproduction number	Lineage-closure model
Life Model IV — Epigenetic Memory & Differentiation	Models current residue as coupled states of methylation, histones, accessibility, RNA, and protein	Prediction/validation program
Life Model V	Multiple memory eigenmodes and cross-layer transfer	Extension of epigenetic memory
Life Model VI	Noncommuting order effects in cell-fate programming	Differential prediction

Model / Application	Core Mapping	Current Meaning
Life Model VII	Induced state versus clonal selection	Separation of alternative explanations
Life Model VIII	Reversible write-erase-rewrite epigenetic editing	Causal rescue experiment
Life Model IX	Extracellular/niche memory	Distributed residue outside the cell
Life Model X	Autonomous closure threshold for memory, execution, repair, boundary, and selection	Boundary of first life/autonomy
WRRR-Cell 0.1–0.4	Fast program + slow residue + nutrient/substrate/biomass/waste/damage state; Perturb-seq audit	Structure PASS; 0/12 relations confirmed in actual K562 data, overall biology verdict OPEN
Integrated Biology 2.1→3.0→4.0/4.1	Integrates DNA→protein→minimal cell→lineage→origin of life with epigenetic memory→fate in one ledger	Auditable research contract

4.3.1 Proposed Biotechnology Applications

Proposed Application	WRRR Perspective
iPSC quality control	Detect donor residue and incomplete resetting
Precision differentiation	Temporal control that opens and closes fate gates in the proper order
Organoids	Coupling spatial signals with lineage memory
Cell therapy	Maintain therapeutic phenotype and safety after division
Immune-cell engineering	Reset memory/exhaustion states
Cancer	Separate induction from selection in drug-tolerant persisters
Regenerative medicine	Control wound memory and redifferentiation
Biomanufacturing	Manage generational stability of high-production phenotypes

The items in this table are not separately validated WRRR laws. They are directions for biotechnology applications proposed in Life Model IV and Integrated Biology.

4.4 WRRR-AI / Embodied Agents and Survival

Model	Core Mapping	Frozen Status
WRRR-Worm 0.1	C. elegans-inspired 23-node/57-relation sensor-neural-motor loop	Structural execution
WRRR-Worm 0.2–0.4	Delayed memory, food seeking, unseen-world benchmark; memoryless/RNN/GRU comparison	Feasibility supported; general superiority not supported
Ground-Up WRRR 0.5	Agent construction from the Core without a worm prior	Expansion of the AI Domain Profile
Memory-required 0.5–0.7	Separates tasks that genuinely require success/failure reward memory	Memory ownership test

Model	Core Mapping	Frozen Status
Minimal-Calculation Survival 0.8	Prioritizes worst-environment survival and reserve rather than maximum reward	Direct implementation of the minimum-computation principle
Autonomous Reserve, Luck, Inheritance 0.9	Separates external reserve, autonomous reserve, luck/events, and inheritance	Extension of survival robustness
C. elegans-Anchored Minimal Survival Circuit 1.0	Minimal survival circuit closer to C. elegans biology	Biological validation separate
Integrated WRRA Artificial Intelligence 2.0	Integrates minimum computation, memory, reserve, luck, and intergenerational transfer	Integrated AI research lineage

Frozen Lesson for AI

WRRA does not mean an AI “algorithm that performs better.” Equal-budget tests in 0.4 did not confirm a general predictive advantage. From the Core perspective, the result lies in auditable separation of SOURCE/RELATION/STATE/BOUNDARY/RENDERER/OBSERVABLE and in treating minimum computation as a boundary of survival, safety, and required memory/reserve rather than average performance.

4.5 WRRA-Econ 0.1–0.5

The economics application tested, in toy models and real macroeconomic data, whether the effects of past shocks can remain as present residue in trust, fear, debt, sentiment, institutional memory, and related variables.

Layer	Content	Frozen Verdict
Structure	Executable toy model with state, residue, resources, boundaries, accounting, and regimes	STRUCTURAL PASS
Synthetic identification	Recovery on fresh seeds when residue is a genuine causal state in the generator	REPLICATED
Real-data generalization	Universal forecasting gain in post-2015 U.S. macroeconomic data	NOT ESTABLISHED
Crisis specificity	Some directional signal in a crisis-memory window, with deterioration in calm periods	PARTIAL SIGNAL
Measurement/timing	Current sentiment does not beat a 36-month-lag temporal placebo	OPEN

Accordingly, WRRA-Econ is not frozen as a theory that asserts an inevitable hidden residue in every economy. Whether the measured coordinates used to date are the intrinsic owners of residue remains open. Preserving this failure illustrates the fail-closed principle of the Core.

4.6 WRRA-Met / Meteorology and Tropical Cyclones

The meteorological application used forward splits to test whether residue compressed into the present state adds independent information for tropical-cyclone track prediction beyond raw finite history or a nonlinear present-state baseline. The frozen study compared multiple residue timescales using JMA best-track data from 1951–2025.

Stage	Essence	Verdict
0.1 synthetic / structural	Checks recoverability when residue is causal truth in a controlled system	Structure confirmed
Strict forward real-data test	train $\leq$ 2014, validation 2015–2019, test 2020–2025	Leakage prevented
Baselines	Persistence, present-only Ridge/ExtraTrees, raw history, and residue placebo	Fair comparison
Decay sensitivity	Point estimates for several residue families are mostly positive, but no cell is significant	Weak direction
Final interpretation	No statistically established independent predictive effect of tropical-cyclone track residue	OPEN / not established

The central lesson for meteorology is not that adding history automatically makes a model WRRA. A candidate residue must outperform raw history, nonlinear present state, a content-free residue placebo, and a temporal-ownership placebo before it can be called independent residue.

4.7 WRRA-Method / Interpretation–Prediction Boundary

Method	Core Role	Scope of Application
Interpretation–Prediction Boundary v1.0	Separates targets closed by current records from future/unknown-information targets	General methodology
Full Formal System v2.1	Includes Bayes risk, compatible-state diameter, threshold sensitivity, Lyapunov/Jacobian amplification, and record quality	Cross-applied to DNA, cells, neural systems, tropical cyclones, and economics
Quantum–Macro classifier	Distinguishes event law, detector/phenotype threshold, amplification, and distinguishable/persistent/redundant records	Physics boundary
Proof-closure audit	Distinguishes static infinite propositions from future prediction	Mathematical application

4.8 Fields Not Yet Frozen as Official WRRA Applications

The author has conducted separate writing and theoretical research in art, aesthetics, and the humanities, but the currently confirmed research lineage contains no independent application frozen as “WRRA-Art,” or under an equivalent name, that implements the Core in an actual model and validation contract. It is therefore excluded from the official Core 1.0 application registry. It may be added after a separate Domain Profile is created.

5. Domain-by-Domain Core Mapping

Domain	Execution Mapping (SOURCE/RELATION/STATE/BOUNDARY)	COMMON CARRIER	PHENOTYPE / RECORD
Physics/cosmology	Dormant/field SOURCE; laws and interactions; quantum/field state and hidden correlations; spacetime and observational boundary	Common transport/carrier	Particles, forces, vacuum, macroscopic Reality / energy, event, and history record

Domain	Execution Mapping (SOURCE/RELATION/STATE/BOUNDARY)	COMMON CARRIER	PHENOTYPE / RECORD
Biology	DNA, molecular stock, environmental input; reaction, regulation, lineage; cellular state and epigenetic residue; membrane, tissue, and generational boundary	Translation, metabolism, membrane transport, and partitioning	Protein, cellular phenotype, fate / molecular ledger and lineage record
AI	Sensation, energy, reward; sparse neural/action relations; activation, adaptation, memory, reserve; body/environment/task boundary	Sensor–state–action network	Action and survival state / resource and survival history
Economics	Shock, policy, flow; credit, price, institutional relations; candidate trust/fear/debt/sentiment residue; market, institutional, regime boundary	Transaction, price, credit, and information network	Price, transaction, crisis/calm state / balance and regime record
Meteorology	Current atmosphere/cyclone state; physical flows and interactions; candidate compressed history/residue; analysis-region/environment boundary	Atmospheric-state transport	Track, intensity, structure / best-track and history ledger
Mathematics/method	Problem, axioms, input; definitions and operations; current computation and proof record; domain, precision, and proof scope	Transformation/inference rule	Computed result and proof state / proof and computation ledger

6. Minimum Conformance Conditions for the WRR Name

- Do not identify SOURCE with OBSERVABLE/PHENOTYPE.
- When effects of the past are required, declare their owner in the current STATE/RESIDUE/RECORD.
- Do not assume that more relations always make a better model. If a null or simpler baseline wins, reduce the WRR extension.
- Where possible, reuse a common carrier across channels without erasing channel identity or ownership.
- The objective of minimum computation is not peak performance but the minimally sufficient structure that satisfies the declared viability/expression contract.
- Include the BOUNDARY and external inflow/outflow in the ledger. Do not enforce internal-only conservation on an open system.
- Do not omit the RENDERER and call an internal variable a phenotype/observable directly.
- Separate the contracts of interpretation, computation, prediction, control, and validation; leave insufficiently supported claims OPEN.
- Do not elevate domain-specific equations, constants, or data to universal laws of the Core.

6.1 Core Conformance Checklist

ID	Question	Verdict
C1	Is SOURCE specified?	No → WRRR Profile not established
C2	Are the owners of current STATE and past RESIDUE/RECORD distinguished?	No → memory claim unclear
C3	Are the BOUNDARY and external inputs specified?	No → ledger/causality incomplete
C4	Does the common carrier specify what is shared and what remains distinct?	No → carrier axis unmet
C5	Does UPDATE generate the next present?	No → not an execution model
C6	Are RENDERER and PHENOTYPE defined?	No → phenotype axis unmet
C7	Is there a minimum-computation comparator or null/baseline?	No → minimality claim unavailable
C8	Are OBSERVABLE/RECORD/LEDGER specified?	No → validation contract not established
C9	Are failure conditions declared in advance?	No → violates fail-closed principle

7. Version Control and Future Extensions

7.1 Changes That Do Not Alter Core 1.0

- Reinforcing MCC 2.1 with string-theoretic microphysical materials and discussing 3+1/10D implementation
- A change in the verdict for a Life Model or WRRR-Cell after new biological data
- Adding a new agent, memory architecture, or reserve mechanism in WRRR-AI
- Using improved residue measurements or null baselines in WRRR-Econ/Met
- Mapping the Core to a new domain such as art, society, law, or language

7.2 Changes That Require Review of Core 1.x or 2.0

- Replacing “minimum sufficiency” with another highest-level objective in the definition of minimum computation
- Removing the common carrier from the required axes or replacing it with independent channel-by-channel generation
- Removing the phenotype/renderer distinction and redefining internal state as observable
- Abandoning the fixed-present principle and adopting a past event list as direct ontology
- Ceasing to use ownership, ledger, or fail-closed principles as Core rules

7.3 Standard Template for a New Domain Profile

Field	Required Entry
Domain name / version	Example: WRRR-Art 0.1
Target contract	Which of interpretation/computation/prediction/control/OPEN?
SOURCE	Current independent inputs and resources
RELATION / LAW	Interactions and rules
STATE / RESIDUE	Current state and effects of the past
BOUNDARY	Inside/outside and inflow/outflow
COMMON CARRIER	Shared transport/update scaffold
UPDATE	Equation generating the next present
RENDERER / PHENOTYPE	Internal-to-external expression
OBSERVABLE / LEDGER	Measurements, conservation/accounting, and records
Minimality baseline	Simpler null/reference model
Failure conditions	Results that would discard or reduce the WRRR extension

8. Final Freeze Declaration

WRRR Core 1.0 began in cosmology, but it is not cosmology itself. It does not declare any particular physical law, prime number, string, DNA sequence, neural network, price, or tropical cyclone to be the Core. What it preserves is a common grammar for auditing how reality or a complex system executes the present, what it transports, what it expresses, and what it records.

Within this common grammar, minimum computation locates the minimally sufficient boundary between deficiency and excess. The common carrier reuses transport across different channels without flattening away ownership. Phenotype separates internal possibility or SOURCE from observed reality. The fixed-present principle converts the past into residue/record remaining in the present; ownership and ledger fix who owns what; and the fail-closed principle blocks claims that cross the boundary among explanation, prediction, and validation.

$$\text{WRRR Core 1.0} = \{\text{Minimum Viable Computation, Common Carrier, Phenotype, Fixed Present, Ownership, Boundary, Renderer, Record/Ledger, Fail-Closed Claims}\} \quad (8.1)$$

Freeze Statement

As of 7 September 2026, WRRR Core 1.0 is frozen under the definitions in this document. All subsequent WRRR applications are treated as Domain Profiles of Core 1.0; the success, failure, or version change of a domain implementation does not automatically require a change to the Core. The Core changes only when one of the three constitutional invariants or the required execution grammar itself changes.

Appendix A. Application Registry Summary

Registry	Scope	Representative Version	Current Status
WRRR-Physics / MCC	Cosmos, forces, matter, quantum phenomena, vacuum, black holes, condensed matter, and nuclei	MCC 2.0/2.1 and Physics series	Active
WRRR-Math / WJNS	Arithmetic SOURCE/RELATION and RH/proof boundary	WRRR 2.0 upstream branch	Research line
WRRR-Bio	DNA→protein→cell→lineage→first life; epigenetic memory/fate	Life 0, I–X; Integrated Biology 4.x	Active
WRRR-Cell	Cellular present residue / gene perturbation	0.1–0.4	OPEN / falsification-oriented
WRRR-AI	Minimum computation, memory, reserve, survival, luck, and inheritance	Integrated AI 2.0	Active
WRRR-Worm	C. elegans-inspired embodied agent	0.1–1.0 lineage	Feasibility; superiority not established
WRRR-Econ	Animal spirits / economic residue	0.1–0.5	Partial signal / measurement OPEN
WRRR-Met	Tropical-cyclone residue / track prediction	0.1–0.5 research line	Effect not established
WRRR-Method	Interpretation–prediction–computation boundary	v1.0 / v2.1	General methodology
WRRR-Art	—	No official Domain Profile	Not frozen

Appendix B. Representative Frozen Documents in the Application Lineage

Minimal Computing Cosmology 2.0 / WRRR-Physics final architecture.

WRRR 2.0: Prime-Port Fixed-Present Relational Architecture — upstream redefinition and downstream impact.

WRRR Integrated Biology 4.0/4.1 — DNA, protein execution, minimal cells, lineage, origin of life, epigenetic memory and cell differentiation.

WRRR Life Model 0 — The First Living System in WRRR.

WRRR-Cell 0.1–0.4 Integrated Research Paper.

WRRR Artificial Intelligence 2.0 — Minimal Computation, Memory, and Survival; WRRR-Worm 0.1–1.0 lineage.

WRRR-Econ 0.1–0.5 Freeze Report.

WRRR-Met Typhoon Track Residue Hypothesis frozen research note.

WRRR Interpretation–Prediction Boundary v1.0 / Full Formal System v2.1.

WRRR Physics notes on antimatter, vacuum/pair processes, superfluidity, nuclear stability/decay and related physicalization studies.